

Lecture 23-24

Chapter 6: Random Vectors and Gaussian Random Variables

Math 4441: Probability and Statistics

Reference: Goodman & Yates – Introduction to Probability and Stochastic Process, 3rd Edition

Vector Notations

Definition 5.2. *Random Vector.* A random vector is a column vector $\mathbf{X} = [X_1, X_2, \dots, X_n]^T$, where each X_i is a random variable.

A random variable is a random vector with $n = 1$. The sample values of the components of a random vector constitute a column vector as well.

Definition 5.3. *Sample-Value Vector.* The sample values of the components of a random vector is a column vector, represented by the bold-face small letter $\mathbf{x} = [x_1, x_2, \dots, x_n]^T$, is defined as the sample-value vector. The i -th component of the vector $\mathbf{x}$, x_i , is a sample value of the random variable X_i . □

Probability Models of Random Vectors

Definition 5.4. *CDF of a Random Vector.* The CDF of a random vector $\mathbf{X}$, for the sample-value vector $\mathbf{x}$, denoted as $F_{\mathbf{X}}(\mathbf{x})$, is given by

$$\begin{aligned} F_{\mathbf{X}}(\mathbf{x}) &= P[\mathbf{X} \leq \mathbf{x}] \\ &= F_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) \\ &= P[X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n] \end{aligned} \tag{5.24}$$

Definition 5.5. (*PMF of a Random Vector:*) The PMF of a discrete random vector $\mathbf{X}$, for the sample-value vector $\mathbf{x}$, denoted as $P_{\mathbf{X}}(\mathbf{x})$, is given by

$$\begin{aligned} P_{\mathbf{X}}(\mathbf{x}) &= P[\mathbf{X} = \mathbf{x}] \\ &= P_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) \\ &= P[X_1 = x_1, X_2 = x_2, \dots, X_n = x_n] \end{aligned} \tag{5.25}$$

Definition 5.6. *PDF of a random vector:* The PDF of a continuous random vector $\mathbf{X}$, for the sample-value vector $\mathbf{x}$, is the multivariate joint PDF of the component random variables $X_1, X_2, \dots, X_n$, and is given by

$$f_{\mathbf{X}}(\mathbf{x}) = f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n). \quad (5.26)$$

Joint Probability Models of Two Random Vectors

Definition 5.7. *Joint PMF of two Random Vectors.* The joint PMF of two discrete random vectors, X with n components and Y with m components, $P_{\mathbf{X},\mathbf{Y}}(\mathbf{x},\mathbf{y})$, is the multivariate joint PMF of $(n + m)$ discrete random variables and is given by

$$P_{\mathbf{X},\mathbf{Y}}(\mathbf{x},\mathbf{y}) = P_{X_1,\dots,X_n,Y_1,\dots,Y_m}(x_1,\dots,x_n,y_1,\dots,y_m). \quad (5.27)$$

Definition 5.8. *Joint PDF of two Random Vectors.* The joint PDF of two continuous random vectors, X with n components and Y with m components, $f_{\mathbf{X},\mathbf{Y}}(\mathbf{x},\mathbf{y})$, is the multivariate joint PMF of $(n + m)$ continuous random variables and is given by

$$f_{\mathbf{X},\mathbf{Y}}(\mathbf{x},\mathbf{y}) = f_{X_1,\dots,X_n,Y_1,\dots,Y_m}(x_1,\dots,x_n,y_1,\dots,y_m). \quad (5.28)$$

Definition 5.9. *Joint CDF of two Random Vectors.* The joint CDF of two discrete/continuous random vectors, X with n components and Y with m components, $F_{\mathbf{X}, \mathbf{Y}}(\mathbf{x}, \mathbf{y})$, is the multivariate joint CDF of $(n + m)$ continuous random variables and is given by

$$F_{\mathbf{X}, \mathbf{Y}}(\mathbf{x}, \mathbf{y}) = F_{X_1, \dots, X_n, Y_1, \dots, Y_m}(x_1, \dots, x_n, y_1, \dots, y_m). \quad (5.29)$$

Example 5.6. Random vector $\mathbf{X}$ has the PDF

$$f_{\mathbf{X}}(\mathbf{x}) = \begin{cases} 6e^{-\mathbf{a}^T \mathbf{x}}, & \mathbf{x} > 0 \\ 0, & \text{otherwise.} \end{cases}$$

where $\mathbf{a}^T = [1 \quad 2 \quad 3]$. What is the CDF of $\mathbf{X}$.

Solution. Because $\mathbf{a}$ has three components, we assume that $\mathbf{X}$ is also a three dimensional vector. Now,

$$\mathbf{a}^T \mathbf{x} = x_1 + 2x_2 + 3x_3, \tag{5.30}$$

and the PDF can be given by

$$f_{\mathbf{X}}(\mathbf{x}) = 6e^{-x_1-2x_2-3x_3}, \quad x_1, x_2, x_3 > 0 \tag{5.31}$$

Finally, the CDF of the vector $\mathbf{X}$ can be found by integrating the PDF of the vector $\mathbf{X}$ over x_1, x_2 and x_3 .

$$\begin{aligned}
 F_{\mathbf{X}}(\mathbf{x}) &= \int_{-\infty}^{x_3} \int_{-\infty}^{x_2} \int_{-\infty}^{x_1} 6e^{-x_1-2x_2-3x_3} dx_1 dx_2 dx_3 \\
 &= \int_0^{x_3} \int_0^{x_2} \int_0^{x_1} 6e^{-x_1} e^{-2x_2} e^{-3x_3} dx_1 dx_2 dx_3 \\
 &= \int_0^{x_3} \int_0^{x_2} 6e^{-2x_2} e^{-3x_3} \left[-e^{-x_1}\right]_0^{x_1} dx_2 dx_3 \\
 &= 6(1 - e^{-x_1}) \int_0^{x_3} e^{-3x_3} \left[-\frac{1}{2}e^{-2x_2}\right]_0^{x_2} dx_3 \\
 &= (1 - e^{-x_1})(1 - e^{-2x_2})(1 - e^{-3x_3})
 \end{aligned}$$

Independent Random Vectors

Two random vectors $\mathbf{X}$ and $\mathbf{Y}$ are independent, if for two discrete random vectors, the following is true

$$P_{\mathbf{X},\mathbf{Y}}(\mathbf{x},\mathbf{y}) = P_{\mathbf{X}}(\mathbf{x})P_{\mathbf{Y}}(\mathbf{y}),$$

and for two continuous random vectors, the following relation is true

$$f_{\mathbf{X},\mathbf{Y}}(\mathbf{x},\mathbf{y}) = f_{\mathbf{X}}(\mathbf{x})f_{\mathbf{Y}}(\mathbf{y}).$$

Example 5.7. Random variables $Y_1, Y_2, \dots, Y_4$ have the following multivariate joint PDF:

$$f_{Y_1, Y_2, \dots, Y_4}(y_1, y_2, \dots, y_n) = \begin{cases} 4, & \text{for } 0 \leq y_1 \leq y_2 \leq 1, 0 \leq y_3 \leq y_4 \leq 1 \\ 0, & \text{otherwise.} \end{cases}$$

Let $\mathbf{V} = [Y_1 \ Y_4]'$ and $\mathbf{W} = [Y_2 \ Y_3]'$. Are $\mathbf{V}$ and $\mathbf{W}$ independent random vectors?

$\mathbf{V}$ are $V_1 = Y_1$, and $V_2 = Y_4$.

$$P_{\mathbf{V}, \mathbf{W}}(v, w) = \begin{cases} 4, & \text{for } 0 \leq v_1 \leq w_1 \leq 1, 0 \leq w_2 \leq v_2 \leq 1 \\ 0, & \text{otherwise.} \end{cases}$$

$$f_{\mathbf{V}}(\mathbf{v}) = \begin{cases} 4(1 - v_1)v_2 & 0 \leq v_1, v_2 \leq 1, \\ 0 & \text{otherwise,} \end{cases}$$

$$f_{\mathbf{W}}(\mathbf{w}) = \begin{cases} 4w_1(1 - w_2) & 0 \leq w_1, w_2 \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

$$f_{\mathbf{V}}(\mathbf{v}) f_{\mathbf{W}}(\mathbf{w}) = \begin{cases} 16(1 - v_1)v_2w_1(1 - w_2) & 0 \leq v_1, v_2, w_1, w_2 \leq 1, \\ 0 & \text{otherwise,} \end{cases}$$

V and W are not independent

Expected Value Vector and Correlation Matrix

Definition 5.10. *Expected value vector:* The expected value vector is the expected value of a random vector and its i -th component is the expected value of the i -th component random variable of the random vector.

If $\mathbf{X}$ is a random vector with n component random variables, $X_1, X_2, \dots, X_n$, then the expected value vector of $\mathbf{X}$, $E[\mathbf{X}]$, is given by

$$E[\mathbf{X}] = \mu_{\mathbf{X}} = \begin{bmatrix} E[X_1] & E[X_2] & \cdots & E[X_n] \end{bmatrix}^T \quad (5.36)$$

For example, the expected value vector of the random vector, $\mathbf{X} = [X_1 \ X_2 \ X_3]^T$ is given by

$$E[\mathbf{X}] = \begin{bmatrix} E[X_1] & E[X_2] & E[X_3] \end{bmatrix}^T. \quad (5.37)$$

Covariance of Random

$$\text{Cov}[\mathbf{X}] = E[(\mathbf{X} - \mu_{\mathbf{X}})(\mathbf{X} - \mu_{\mathbf{X}})^T].$$

$$\begin{aligned} Y_{i,j} &= E[(X_i - E[X_i])(X_j - E[X_j])] \\ &= \text{Cov}[X_i, X_j]. \end{aligned}$$

Definition 5.11. *Covariance Matrix.* The covariance matrix of a random vector $\mathbf{X}$, $Cov[\mathbf{X}]$, represents covariance of every pair of components random variables of the random vector, and is given by

$$Cov[\mathbf{X}] = \begin{bmatrix} \text{Var}[X] & Cov[X_1, X_2] & \cdots & Cov[X_1, X_n] \\ Cov[X_2, X_1] & Var[X_2, X_2] & \cdots & Cov[X_2, X_n] \\ \vdots & \vdots & \ddots & \vdots \\ Cov[X_n, X_1] & Cov[X_n, X_2] & \cdots & Var[X_n, X_n] \end{bmatrix}$$

$$\begin{aligned}
\text{Cov}[\mathbf{X}] &= E[(\mathbf{X} - \mu_{\mathbf{X}})(\mathbf{X} - \mu_{\mathbf{X}})^T] \\
&= E[\mathbf{X}\mathbf{X}^T - \mathbf{X}\mu_{\mathbf{X}}^T - \mu_{\mathbf{X}}\mathbf{X}^T + \mu_{\mathbf{X}}\mu_{\mathbf{X}}^T] \\
&= E[\mathbf{X}\mathbf{X}^T] - E[\mathbf{X}\mu_{\mathbf{X}}^T] - E[\mu_{\mathbf{X}}\mathbf{X}^T] + E[\mu_{\mathbf{X}}\mu_{\mathbf{X}}^T] \\
&= E[\mathbf{X}\mathbf{X}^T] - E[\mathbf{X}]\mu_{\mathbf{X}}^T - \mu_{\mathbf{X}}E[\mathbf{X}^T] + \mu_{\mathbf{X}}\mu_{\mathbf{X}}^T \\
&= E[\mathbf{X}\mathbf{X}^T] - \mu_{\mathbf{X}}\mu_{\mathbf{X}}^T - \mu_{\mathbf{X}}\mu_{\mathbf{X}}^T + \mu_{\mathbf{X}}\mu_{\mathbf{X}}^T \\
&= E[\mathbf{X}\mathbf{X}^T] - \mu_{\mathbf{X}}\mu_{\mathbf{X}}^T
\end{aligned}$$

Correlation Matrix

Definition 5.12. *Correlation Matrix:* The correlation of a random vector $\mathbf{X}$ with n components is an $n \times n$ matrix, $R_{\mathbf{X}}$, whose (i, j) -th elements $R_{\mathbf{X}}(i, j) = E[X_i, X_j]$, and is given by

$$R_{\mathbf{X}} = \begin{bmatrix} E[X_1^2] & E[X_1, X_2] & \cdots & E[X_1, X_n] \\ E[X_2, X_1] & E[X_2^2] & \cdots & E[X_2, X_n] \\ \vdots & \vdots & \ddots & \vdots \\ E[X_n, X_1] & E[X_n, X_2] & \cdots & E[X_n^2] \end{bmatrix}$$

Finally, the covariance of a random vector $\mathbf{X}$ in terms of its correlation $R_{\mathbf{X}}$ and expectation $\mu_{\mathbf{X}}$ is given by

$$Cov[\mathbf{X}] = R_{\mathbf{X}} - \mu_{\mathbf{X}}\mu_{\mathbf{X}}^T.$$



Example 5.8. The two dimensional random vector $\mathbf{X} = [X_1 \ X_2]^T$ has the following PDF:

$$f_{\mathbf{X}}(\mathbf{x}) = \begin{cases} 2, & \text{for } 0 \leq x_1 \leq x_2 \leq 1 \\ 0, & \text{otherwise.} \end{cases}$$

Find the expected value $E[\mathbf{X}]$, correlation matrix $R_{\mathbf{X}}$, and coariance matrix $\text{Cov}_{\mathbf{X}}$ for the random vector $\mathbf{X}$.

$$\begin{aligned}f_{X_1}(x_1) &= \int_{x_1}^1 2dx_2 \\&= 2(1-x_1), \quad 0 \leq x_1 \leq 1.\end{aligned}$$

$$\begin{aligned}f_{X_2}(x_2) &= \int_0^{x_2} 2dx_1 \\&= 2x_2, \quad 0 \leq x_2 \leq 1.\end{aligned}$$

The expected values of the component random variables are given by

$$\begin{aligned} E[X_1] &= \int_0^1 x_1 \cdot 2(1 - x_1) dx_1 \\ &= 2 \left[\frac{x_1^2}{2} \Big|_0^1 - \frac{x_1^3}{3} \Big|_0^1 \right] \\ &= 2 \left(\frac{1}{2} - \frac{1}{3} \right) \\ &= 2 \times \frac{3 - 2}{6} \\ &= \frac{1}{3}. \end{aligned}$$

$$\begin{aligned}
 E[X_2] &= \int_0^1 x_2 \cdot 2x_2 dx_2 \\
 &= 2 \frac{x_2^3}{3} \Big|_0^1 \\
 &= \frac{2}{3}.
 \end{aligned}$$

Therefore, the expected value vector for $\mathbf{X}$ is

$$E[\mathbf{X}] = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} \end{bmatrix}^T$$

Now the second moments of the component random variables are given by

$$\begin{aligned} E[X_1^2] &= \int_0^1 x_1^2 \cdot 2(1 - x_1) dx_1 \\ &= 2 \left[\frac{x_1^3}{3} \Big|_0^1 - \frac{x_1^4}{4} \Big|_0^1 \right] \\ &= 2 \left(\frac{1}{3} - \frac{1}{4} \right) \\ &= 2 \times \frac{4 - 3}{12} \\ &= \frac{1}{6}. \end{aligned}$$

$$\begin{aligned}
 E[X_2^2] &= \int_0^1 x_2^2 \cdot 2x_2 dx_2 \\
 &= 2 \frac{x_2^4}{4} \Big|_0^1 \\
 &= \frac{1}{2}.
 \end{aligned}$$

And the expected value of the product of the two component random variables is given by

$$\begin{aligned} E[X_1X_2] &= \int_0^1 \int_0^{x_2} x_1x_2 2dx_1dx_2 \\ &= \int_0^1 x_2.\frac{x_1^2}{2}\Big|_0^{x_2} dx_2 \\ &= \int_0^1 x_2.x_2^2dx_2 \\ &= \frac{x_2^4}{4}\Big|_0^1 \\ &= \frac{1}{4}. \end{aligned}$$

Therefore, the correlation matrix for the random vector $\mathbf{X}$ is

$$R_{\mathbf{X}} = \begin{bmatrix} \frac{1}{6} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} \end{bmatrix}.$$

Finally, the covariance matrix of the random vector $\mathbf{X}$ is given by

$$\begin{aligned} \text{Cov}_{\mathbf{X}} &= R_{\mathbf{X}} - \mu_{\mathbf{X}}\mu_{\mathbf{X}}^T \\ &= \begin{bmatrix} \frac{1}{6} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} \end{bmatrix} - \begin{bmatrix} \frac{1}{3} \\ \frac{2}{3} \end{bmatrix} \begin{bmatrix} \frac{1}{3} & \frac{2}{3} \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{6} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} \end{bmatrix} - \begin{bmatrix} \frac{1}{9} & \frac{2}{9} \\ \frac{2}{9} & \frac{4}{9} \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{18} & \frac{1}{36} \\ \frac{1}{36} & \frac{1}{18} \end{bmatrix}. \end{aligned}$$

Cross-correlation and Cross-covariance of Random Vectors

The cross-correlation of a pair of random vectors $\mathbf{X}$ with n component random variables and $\mathbf{Y}$ with m component random variables is an $n \times m$ matrix $R_{\mathbf{X}, \mathbf{Y}}$ with the (i, j) -th element $R_{\mathbf{X}, \mathbf{Y}}(i, j) = E[X_i Y_j]$ which, in the vector form, is given by

$$\mathbf{R}_{\mathbf{X}\mathbf{Y}} = E[\mathbf{X}\mathbf{Y}^T].$$

Definition 5.13. *Cross-covariance:* The cross-covariance of a pair of random vectors $\mathbf{X}$ with n component random variables and $\mathbf{Y}$ with m component random variables is an $n \times m$ matrix $\mathbf{Cov}[\mathbf{X}, \mathbf{Y}]$ with the (i, j) -th element $\mathbf{Cov}[\mathbf{X}, \mathbf{Y}](i, j) = \mathbf{Cov}[X_i Y_j]$ which, in the vector form, is given by

$$\mathbf{Cov}[\mathbf{X}, \mathbf{Y}] = E[(\mathbf{X} - \boldsymbol{\mu}_{\mathbf{X}})(\mathbf{Y} - \boldsymbol{\mu}_{\mathbf{Y}})^T].$$

$\mathbf{X}$ is an n -dimensional random vector with expected value $\boldsymbol{\mu}_{\mathbf{X}}$, correlation $\mathbf{R}_{\mathbf{X}}$, and covariance $\mathbf{C}_{\mathbf{X}}$. The m -dimensional random vector $\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$, where $\mathbf{A}$ is an $m \times n$ matrix and $\mathbf{b}$ is an m -dimensional vector, has expected value $\boldsymbol{\mu}_{\mathbf{Y}}$, correlation matrix $\mathbf{R}_{\mathbf{Y}}$, and covariance matrix $\mathbf{C}_{\mathbf{Y}}$ given by

$$\boldsymbol{\mu}_{\mathbf{Y}} = \mathbf{A}\boldsymbol{\mu}_{\mathbf{X}} + \mathbf{b},$$

$$\mathbf{R}_{\mathbf{Y}} = \mathbf{A}\mathbf{R}_{\mathbf{X}}\mathbf{A}' + (\mathbf{A}\boldsymbol{\mu}_{\mathbf{X}})\mathbf{b}' + \mathbf{b}(\mathbf{A}\boldsymbol{\mu}_{\mathbf{X}})' + \mathbf{b}\mathbf{b}',$$

$$\mathbf{C}_{\mathbf{Y}} = \mathbf{A}\mathbf{C}_{\mathbf{X}}\mathbf{A}'.$$

Example 5.9. Given the expected value $\boldsymbol{\mu}_{\mathbf{X}}$, the correlation $\mathbf{R}_{\mathbf{X}}$, and the covariance $\mathbf{Cov}[\mathbf{X}]$ of random vector $\mathbf{X}$ in Example 5.8, and $\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$, where

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 6 & 3 \\ 3 & 6 \end{bmatrix}$$

and

$$\mathbf{b} = \begin{bmatrix} 0 \\ -2 \\ -2 \end{bmatrix}.$$

Find the expected value $\boldsymbol{\mu}_{\mathbf{Y}}$, the correlation $\mathbf{R}_{\mathbf{Y}}$, and the covariance $\mathbf{Cov}[\mathbf{Y}]$.

$$\boldsymbol{\mu}_Y = \mathbf{A}\boldsymbol{\mu}_X + \mathbf{b}$$

$$= \begin{bmatrix} 1 & 0 \\ 6 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} \frac{1}{3} \\ \frac{2}{3} \end{bmatrix} + \begin{bmatrix} 0 \\ -2 \\ -2 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{3} \\ 4 \\ 5 \end{bmatrix} + \begin{bmatrix} 0 \\ -2 \\ -2 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{3} \\ 2 \\ 3 \end{bmatrix}.$$

The correlation $\mathbf{R}_\mathbf{X}$ is given by

$$\begin{aligned}
 \mathbf{R}_\mathbf{Y} &= \mathbf{A}\mathbf{R}_\mathbf{X}\mathbf{A}^T + (\mathbf{A}\boldsymbol{\mu}_\mathbf{X})\mathbf{b}^T + \mathbf{b}(\mathbf{A}\boldsymbol{\mu}_\mathbf{X})^T + \mathbf{b}\mathbf{b}^T \\
 &= \begin{bmatrix} 1 & 0 \\ 6 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} \frac{1}{6} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 & 6 & 3 \\ 0 & 3 & 6 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 6 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} \frac{1}{3} \\ \frac{2}{3} \\ \frac{2}{3} \end{bmatrix} \begin{bmatrix} 0 & -2 & -2 \end{bmatrix} + \\
 &= \begin{bmatrix} \frac{1}{3} \\ 4 \\ 5 \end{bmatrix} + \begin{bmatrix} 0 \\ -2 \\ -2 \end{bmatrix} \\
 &= \begin{bmatrix} \frac{1}{6} & \frac{13}{12} & \frac{4}{3} \\ \frac{13}{12} & \frac{15}{2} & \frac{37}{4} \\ \frac{4}{3} & \frac{37}{4} & \frac{25}{2} \end{bmatrix}.
 \end{aligned}$$

$$\text{Cov}[\mathbf{Y}] = \mathbf{A}\text{Cov}[\mathbf{X}]\mathbf{A}^T$$

$$= \begin{bmatrix} 1 & 0 \\ 6 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} \frac{1}{18} & \frac{1}{36} \\ \frac{1}{36} & \frac{1}{18} \end{bmatrix} \begin{bmatrix} 1 & 6 & 3 \\ 0 & 3 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{18} & \frac{1}{36} \\ \frac{5}{12} & \frac{1}{3} \\ \frac{1}{3} & \frac{5}{12} \end{bmatrix} \begin{bmatrix} 1 & 6 & 3 \\ 0 & 3 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{18} & \frac{5}{12} & \frac{1}{3} \\ \frac{5}{12} & \frac{7}{2} & \frac{13}{4} \\ \frac{1}{3} & \frac{13}{4} & \frac{7}{2} \end{bmatrix}.$$

If $\mathbf{X}$ is a continuous random vector and $\mathbf{A}$ is an invertible matrix, then $\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$ has PDF

$$f_{\mathbf{Y}}(\mathbf{y}) = \frac{1}{|\det(\mathbf{A})|} f_{\mathbf{X}}(\mathbf{A}^{-1}(\mathbf{y} - \mathbf{b}))$$

The vectors $\mathbf{X}$ and $\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$ have cross-correlation $\mathbf{R}_{\mathbf{XY}}$ and cross-covariance $\mathbf{C}_{\mathbf{XY}}$ given by

$$\mathbf{R}_{\mathbf{XY}} = \mathbf{R}_{\mathbf{X}}\mathbf{A}' + \mu_{\mathbf{X}}\mathbf{b}',$$

$$\mathbf{C}_{\mathbf{XY}} = \mathbf{C}_{\mathbf{X}}\mathbf{A}'.$$

Example 8.9

Continuing Example 8.8 for random vectors $\mathbf{X}$ and $\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$, calculate

- (a) The cross-correlation matrix $\mathbf{R}_{\mathbf{XY}}$ and the cross-covariance matrix $\mathbf{C}_{\mathbf{XY}}$.
- (b) The correlation coefficients ρ_{Y_1, Y_3} and ρ_{X_2, Y_1} .

(a) Direct matrix calculation using Theorem 8.9 yields

$$\mathbf{R}_{\mathbf{XY}} = \begin{bmatrix} 1/6 & 13/12 & 4/3 \\ 1/4 & 5/3 & 29/12 \end{bmatrix}; \quad \mathbf{C}_{\mathbf{XY}} = \begin{bmatrix} 1/18 & 5/12 & 1/3 \\ 1/36 & 1/3 & 5/12 \end{bmatrix}.$$

(b) Referring to Definition 5.6 and recognizing that $\text{Var}[Y_i] = C_Y(i, i)$, we have

$$\rho_{Y_1, Y_3} = \frac{\text{Cov}[Y_1, Y_3]}{\sqrt{\text{Var}[Y_1] \text{Var}[Y_3]}} = \frac{C_Y(1, 3)}{\sqrt{C_Y(1, 1)C_Y(3, 3)}} = 0.756 \quad (8.38)$$

Similarly,

$$\rho_{X_2, Y_1} = \frac{\text{Cov}[X_2, Y_1]}{\sqrt{\text{Var}[X_2] \text{Var}[Y_1]}} = \frac{C_{\mathbf{XY}}(2, 1)}{\sqrt{C_{\mathbf{X}}(2, 2)C_{\mathbf{Y}}(1, 1)}} = 1/2. \quad (8.39)$$